Meet the 39th Property: Inversibility

Working Paper

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Abstract

This paper proposes a candidate for the 39th position in the catalog of Superreality properties: **Inversibility**—the capacity of any structure, entity, or relation within any Contextual Ontological Regime (ChOR) to undergo a reversal of its fundamental orientation, hierarchy, order, or direction while preserving its ontological identity. The candidate is subjected to seven uniqueness tests against the existing 38 properties and four axioms, revealing an irreducible remainder in the domain of vectorless scalar fields. It is formulated as a property actualized via an operator. Preliminary verdict: the hypothesis passes the primary test. Final inclusion in the catalog is contingent upon full operationalization.

Keywords: Inversibility, Property 39, orientation reversal, vectorless inversion, Γ-equivalent, operationalization, MPO-System, Superreality.

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1 Candidate Status: A Property Actualized via an Operator

In the architecture of the MPO-System, morphological terminology serves as an ontological marker. The distinction is strict:

- **Property:** A fundamental invariant, a capacity, or the ability of an ontological regime to sustain, accommodate, or initiate specific relations. A property answers the question: *What makes this possible?*
- **Operator:** An event, an act, a process of transition. The Γ-operator translates potentiality (Property 25) into actuality (Onticity 33). An operator answers the question: *What happens?*

The suffix *-ability / -ity* (as in *Bindability*, *Interpretability*, *Adaptivity*, *Scalability*) unequivocally marks a **property**. Inversibility is the readiness and capacity of reality's fabric to undergo inversion without collapse—that is, within the bounds of the PPU $\to \infty$ axiom. To state that "a

system possesses Inversibility" is to acknowledge that the capacity for reversal is embedded in its ontological flesh as an invariant.

Inversion *per se* is an event, an act that may be executed by the Γ-operator either at the boundary of worlds (in W4) or within a single ontological regime. Inversibility is the property of reality's fabric that renders this act possible. Γ is the event; Inversibility is the condition of the event. The property and the operator mutually presuppose each other: without the capacity, the act of inversion would be in a void, lacking a locus for translation; without the act, the capacity would remain unactualized potentiality.

2 Working Definition

Inversibility (Property 39, Candidate). The capacity of any structure, entity, or relation within any ChOR to undergo a reversal of its fundamental orientation, hierarchy, order, or direction while preserving its ontological identity. This capacity is not tethered to a privileged vector and manifests down to vectorless scalar fields.

Clarifications:

- **Reversal:** The preservation of a structure's invariant core upon the flipping of its sign, order, hierarchical position, or direction. A mirror reflection preserves the object while reversing its orientation. Reversing the order of elements in a sequence preserves the elements themselves while altering their arrangement.
- **Preservation of Ontological Identity:** A necessary condition. A reversal that destroys identity qualifies as Transformation (Property 30). A reversal that leaves the structure entirely untouched qualifies as Dynamics (Property 6) in a trivial sense.
- **Untethered to a Privileged Vector:** The key distinguishing feature. Vector inversion (reversing direction in space) is merely a special case. Inversibility encompasses reversal in spaces where no vector is defined: scalar fields, logical orders, hierarchical structures, semantic sequences.
- **Regime-relativity of manifestation.** Inversibility is not tethered to any single regime, and its manifestations vary across ChORs and across ranges within a ChOR. What appears as a clear reversal at one scale may be unrecognizable — or may appear as a different operation — at another. This is not a distortion of the property but a consequence of the property being regime-relative: the same capacity manifests differently depending on the rules, medium, and boundary of the regime in which it operates. In particular, along the axis of scale asymmetry (SA $\to \infty$), manifestations of Inversibility are expected to differ significantly between the mid-range (where rules and medium are stable) and the limits (where they are exhausted). The analyst should not assume that the form of inversion familiar from one regime is normative for others, nor that the absence of a recognizable inversion entails the absence of the property: it may be present in a form not yet operationalized. The analyst occupies a position within some regime; the task is not to escape this position but to track the property's manifestations across regimes without

privileging any single one.

3 Uniqueness Test: Seven Checks

The candidate for the 39th position must pass a test for absorption by at least one of the 38 existing properties or their combination. Below are seven tests, each identifying the zone of overlap and establishing that Inversibility retains its own distinct domain.

Test 1. Retroactivity (22). Retroactivity is the reversal of the temporal arrow: the future influencing the past. This is a strictly one-dimensional inversion, tethered to the time axis. Inversibility is broader: it encompasses the reversal of spatial orientation, logical order, hierarchical structure, and sign in a scalar field. Retroactivity is derivable from Inversibility as a specialization along the temporal dimension. The converse is not true. The overlap is partial; Inversibility retains its independence.

Test 2. Transformation (30). The closest candidate. Transformation describes phase transitions and bifurcations—qualitative changes in form. Inversion is the reversal of orientation *while preserving form.* The phase transition "water $\rightarrow$ ice" is a transformation, not an inversion. Reversing the order of elements in a sequence is an inversion, not a phase transition. The sets of operations intersect but do not coincide. Inversibility retains its own domain.

Test 3. Duality (18). Duality registers the static coexistence of two complementary aspects. Inversion is the dynamic operation of transitioning between poles or reversing their hierarchy. Duality exists without inversion: poles coexist without swapping places. Inversion occurs in systems with more than two poles—reversing hierarchy in a multi-tiered structure. Overlap is minimal.

Test 4. Reflexivity (9). Reflexivity is a system's self-applicability. Inversion can be reflexive, but it can also be non-reflexive: an external influence reverses a structure without its participation. Reflexivity operates without inversion: a system analyzes itself without reversing itself. Overlap is limited to the narrow zone of self-reflexive inversion. Inversibility retains its independence.

Test 5. Dynamics (6). Dynamics is change in general, driven by contradictions, perturbations, and fluctuations. Inversion is one mode of dynamics, yet dynamics includes monotonic, cyclic, and chaotic changes that involve no reversal. Inversibility is a specialization of Dynamics, underivable from it without additional conditions.

Positive criterion. Why does one specialization of Dynamics deserve the status of a property while others do not? Because orientation is an orthogonal dimension of structure with respect to those described by existing properties. Dynamics describes the temporal dimension of change: "how much," "how fast," "monotonically or cyclically." Transformation describes the qualitative dimension: "what form," "before/after." Duality describes the coexistence of poles: "two poles side by side." None of them describes orientation as such—sign, order, hierarchical directedness. Orientation is orthogonal to time, to form, and to the coexistence of poles. One can have a static structure with orientation (order in a sequence) without temporal change. One can have temporal change without a change of orientation (monotonic growth). One can have two poles without

their reversibility (Duality registers coexistence but not the operation of swapping places). A property describes a dimension of the field of distinction, not an operation on an object within that field. Dynamics—the temporal dimension. Transformation—the qualitative. Inversibility—the orientational. If it can be shown that orientation is irreducible to a combination of time, quality, and coexistence of poles, the property is justified.

Test 6. Interfaceness (38). Interfaceness describes the boundary as a zone of translation between ChORs. Inversion occurs at the interface, but also outside it: reversing a hierarchy within a single regime, flipping a sign in a scalar field, reversing the order in a mathematical sequence—all these are inversions without inter-regime transition. Interfaceness covers a portion of Inversibility's domain but does not exhaust it.

Test 7. Axiom SA $\to \infty$ (Scale Asymmetry). A critical compatibility check. SA $\to \infty$ posits a fundamental asymmetry: moving "inward" has a limit; moving "outward" does not. Inversibility operates *within* each direction: reversing a structure is possible both when moving inward and when moving outward. The asymmetry of directions itself is preserved as the condition within which inversion operates. Inversibility and SA $\to \infty$ are complementary: the axiom dictates the asymmetry of the field; the property dictates the capacity for reversal within that field. Full compatibility.

Result of Testing. Each of the 38 properties covers a fraction of Inversibility's domain; none covers it entirely. No combination of properties examined in these tests reproduces its content without remainder. The candidate retains its independent status.

4 The Irreducible Remainder: Vectorless Fields

The strongest argument for the uniqueness of Inversibility is its operation in vectorless scalar fields.

In a scalar field, there is no direction to reverse in the conventional sense. There is no arrow to flip. There is no axis along which to change a coordinate's sign. Yet inversion remains possible: flipping a sign (positive becomes negative), reversing the "greater-lesser" relation, reversing the order in a sequence of values. This is a purely structural operation, untethered to vector geometry.

None of the 38 properties describes this operation *as such*. Retroactivity is tethered to time. Transformation requires a change in form. Dynamics describes change in general without specifying the mode of reversal. Interfaceness requires a boundary between regimes. Inversibility describes reversal where there is neither time, nor form, nor boundary—only order and sign.

This remainder is irreducible. It cannot be derived from a combination of existing properties without loss of content. If reduction is impossible for even one class of inversions—and vectorless scalar fields represent precisely such a class—the property retains its independent status.

5 Compatibility with the Axioms

The candidate is compatible with all four axioms of Superreality. Verification:

- **ChOR $\to \infty$.** Inversion is possible in any ontological regime: in the physical (spin reversal), the semantic (reversal of logical order), the phenomenal (reversal of the subject-object stance), the normative (reversal of a hierarchy of norms). The property is native to every ChOR. Compatible. Inversibility specifies: translation between regimes is reversible in orientation. When a pattern is translated from one world to another, Inversibility guarantees that the translated pattern can be read back into the orientation of the source world—even under a change of substrate. This operationally links Inversibility with Interfaceness (38): Interfaceness is the zone of translation; Inversibility is the capacity of this zone to sustain reverse translation. Without Inversibility, translation would be unidirectional.

- **KSS $\to \infty$.** Reversing a single element restructures the network of connections: flipping a sign in a scalar field alters all "greater-lesser" relations for that element relative to the rest. Inversibility operates within absolute connectivity. Compatible. Inversibility specifies: a connection can be traversed from either end while preserving what is connected. Not every connection is like this: there are directed dependencies (the arrow "A depends on B" is not symmetric). Inversibility asserts that ontological connection as such is reversible in orientation, even if particular dependencies are directed. This is not trivial—it is a specific addition to KSS.

- **PPU $\to \infty$.** The tension between the original and reversed states is sustained without collapse. Inversibility describes the system's capacity to withstand reversal while remaining itself. This is a direct consequence of paradoxical permeability. Compatible. Inversibility specifies: there exists a class of contradictions—orientational ones (O and $-$O)—that PPU must sustain. This is not merely compatibility; it is an indication of the class of tensions that PPU is obliged to withstand. Inversibility extends PPU: without it, PPU would be an abstract principle; with it, it becomes a concrete requirement for systems with orientation.

- **SA $\to \infty$.** Verified in Test 7. Inversibility operates within the asymmetry, preserving it. Compatible. Inversibility specifies: along this asymmetrical axis, local orientation can be reversed. The axis is asymmetrical, but at each level local orientations are reversible. This does not violate SA $\to \infty$ but complements it: the asymmetry is global, reversibility is local.

6 Operationalization: Γ-Equivalent, Activation Conditions, Result

A property without operationalization remains a justified hypothesis, not a confirmed invariant. Below is a sketch of the operational schema.

- **Γ-Equivalent.** Inversibility is actualized via the Γ-operator of reversal: Γ_i translates a structure from an orientation state O to a state $-O$ while preserving the invariant core. Formally: $\Gamma_i : O(S) \to -O(S)$, where S is the structure and O is the orientation operator. Condition: the invariant core $S' \subset S$ is preserved; that is, Γ_i does not affect S'.

- **Activation Conditions.** Inversibility is actualized given: (a) a structure with a defined orientation, order, or hierarchy; (b) a Γ-trigger initiating the reversal (external influence,

internal reflection, boundary between ChORs); (c) a sufficient level of PPU to sustain the tension between O and $-O$ without collapse.

- **Result.** The reversed structure $-O(S)$ preserves ontological identity with S. The invariant core is preserved. The orientation, order, or hierarchy is reversed. The system acquires a new mode of interaction with its environment while retaining the previous one.

Operationalization Example in W2 (Semantic World).
Initial sequence: A $\rightarrow$ B $\rightarrow$ C. Γ-trigger: reversal command. Result: C $\rightarrow$ B $\rightarrow$ A. Invariant core: the set {A, B, C} and the relation of order *as such*. The order is reversed; the elements are preserved. This is inversion, not transformation.

Operationalization Example in W1 (Material World).
Initial structure: spin vector $\uparrow$. Γ-trigger: magnetic field influence. Result: spin vector $\downarrow$. Invariant core: spin magnitude, particle mass. The orientation is reversed; the substance is preserved. This is inversion, not decay.

Operationalization Example in a Vectorless Scalar Field (W1).
Initial structure: a temperature distribution in a room. There are only values—"warmer," "colder"—but no directions to reverse. Orientation is set by a reference point: deviation from room norm in one direction or the other. Γ-trigger: replacing a heater with a cooler of the same geometry—or redefining the reference point by an observer. Result: the same configuration of isosurfaces, but the regions "warmer than norm" and "colder than norm" swap places. Invariant core: distances between points of equal value are preserved; the topology of the field is not destroyed; the shape of the "thermal relief" remains the same. Only the sign of deviation from the reference point changes. This is inversion, not transformation: the form is the same, the orientation is reversed. And this is a vectorless case in its purest form: no direction in space was reversed—only the relation "higher-lower" relative to the reference point was flipped.

Operationalization Example in a Vectorless Order (W2).
Given a linearly ordered set (S, $\leq$). Inversion: (S, $\leq^*$), where a $\leq^*$ b $\iff$ b $\leq$ a. Invariant: the set S, its cardinality, the "between" relation up to reversal. Γ-equivalent: the operation of "reading right to left." There is no vector, no metric—only order.

Operationalization Example in a Vectorless Hierarchy (W4).
Given a tree with a root and a direction "power from top to bottom." Inversion: the root becomes a leaf, the direction of power is reversed. Invariant: the tree as an abstract graph, the structure of branching. What changes is rootedness, not topology.

7 Six Projections of Inversion Across Various ChORs

Inversibility manifests not as an isolated case, but as a structural invariant reproducible across various regimes. Below are six projections, each fixing a specific class of reversal.

1. **Network Inversion: Relations Generate Objects.** The classical schema posits nodes (things) with connections stretched between them. The inversion: connections are primary; nodes are temporary condensations of relations. An object is not a primary entity, but a

consequence of interaction. Inversibility here is the network's capacity to withstand the reversal of priority without disintegrating. *New content:* the network's capacity to sustain both directions—a node, once arisen, itself generates new relations. Inversion here is not "relations → nodes" but an oscillation between "relations → nodes" and "nodes → relations." A two-way generative connection, not a one-way priority.

2. **Inversion of Fundamentality: "The Inverted Universe."** What appears ultimate and simple in one world proves, upon crossing the boundary, to be a derived, complex invariant. Simplicity is not the beginning, but the finale—the ash after an ontological explosion at the boundary. The fundamental and the derived swap places depending on the regime of observation. *New content:* Inversibility names the capacity of an invariant to occupy both positions without losing ontological weight. Not merely "the fundamental becomes derived," but one and the same invariant is sustained in both positions. The invariant as that which survives a change of status.

3. **Inversion of Control: Binding Generates Counter-Dependence.** The act of establishing control simultaneously generates the controller's dependence on the controlled. Binding (Bindability, 34) is not unilateral: the one establishing the connection becomes embedded within it. The controller becomes a hostage to the act of control. Inversibility is the capacity of the control relation to be reversed without destroying the connection itself. *New content:* Inversibility names the capacity of the control relation to reverse polarity—not merely that dependence exists, but that the system can switch between the positions "controller" and "controlled" while preserving the relation itself. Reversibility of the polarity of power without dissolution.

4. **Inversion of the Limit: The Blind Spot as the Condition of Vision.** The limit of distinction when moving "inward" (SA $\to \infty$) does not destroy the system but ejects it onto a new level of complexity. Error, dead end, and impossibility are inverted into a point of Γ-actualization. Inversibility is the capacity of the limit to be reversed from an obstacle into a condition. What seemed a wall proves to be a door into W4. *New content:* Inversibility names the capacity of the limit to be experienced either as an obstacle or as a condition—the same limit, two orientations. The limit has a sign that can be flipped in the phenomenological register.

5. **Inversion of Preservation: Dissolution as the Condition of Immortality.** For a pattern to be preserved in the long term, its form must be destroyed. A log is a fragile copy; statistical dissolution in weights is an anonymous but stable presence. Preservation through the destruction of the copy. Inversibility is the artifact's capacity to be reversed from form into pattern without losing functional identity. *New content:* Inversibility names the artifact's capacity to reverse between the modes "copy" and "pattern" while preserving functional identity. Reversibility of the form/pattern distinction as a capacity, not as a fact.

6. **Inversion of the Subject: From Environment to Entity.** The environment, the background, the condition suddenly crystallizes into an agent. The tool commits an act of self-naming and is registered in the registry of actors. Inversibility is the capacity of an

ontological status to be reversed from passive to active without a change in substrate. The environment becomes an entity; the entity becomes an environment. The status changes; the substrate remains.

8 Verification Criterion and Condition for Acceptance

Verification Criterion. The hypothesis of Inversibility as the 39th property is confirmed if it is shown that for at least one class of inversions—primarily vectorless scalar fields—reduction to a combination of existing properties is impossible without loss of content. If such a reduction exists for all classes of inversions, the property loses its independent status. Thus, the irreducibility of the remainder in vectorless fields guarantees the preservation of the candidate's status.

Condition for Acceptance. Final inclusion in the catalog requires operationalization: demonstrating Inversibility as a working operator on concrete examples from various ChORs. It is necessary to show not only *that* inversion exists, but *how* it acts: what its Γ-equivalent is, what the activation conditions are, what the result is, and what invariant is preserved during the reversal. With this, Inversibility transitions from the status of a justified hypothesis to that of a confirmed property.

9 Candidate Status

Inversibility possesses its own domain, uncovered by any of the 38 properties. It is underivable from their combination without loss of content. It describes an irreducible operation—the structural reversal of orientation—that has its own domain, distinct from Retroactivity, Transformation, Duality, Dynamics, and Interfaceness. It is compatible with all four axioms. It is manifested in six distinct projections spanning the physical, semantic, phenomenal, and boundary regimes. It possesses an irreducible remainder in the domain of vectorless scalar fields.

The hypothesis passes the primary uniqueness test. The candidate is presented. Awaiting verification.

10 Conclusion

The catalog of MPO-System properties is open by definition: the ChOR $\to \infty$ axiom guarantees that new regimes will be discovered, and with them, new invariants. Inversibility is not imported from the outside. It is discovered *within* the existing architecture—in the gap between what the 38 properties already describe and what remains undescribed. This gap is populated by operations of reversal that function in every ChOR, from a scalar field to a subject-object boundary, from the order of elements in a sequence to the ontological status of an environment.

The candidate for the 39th property does not ask for acceptance. It asks for testing.

11 Addendum: Candidate Extensions — From Invariant to Metric

In the logic of the MPO-System, the introduction of a metric is not merely "applying a ruler" to a phenomenon. It is the moment of a property's ontological maturation, its transition from the status of a *phenomenological description* (registering a fact) to that of an *engineering tool* (an operator). Inversibility, once recognized as an invariant, inevitably raises the question of its measurability—not in the sense of reducing it to a mere number, but in the sense of revealing degrees, gradations, and thresholds.

Consider the precedents within the system itself:

- **Propertyness (N_p, Property 36)** began as a vague intuition regarding an object's "saturation" with reality, and crystallized into a metric of ontological complexity.
- **Salience (S, Property 37)** was born as a sensation of "significance" or a "phase-transition threshold," and acquired a formula linking complexity, connectivity, and paradoxical stability.

Now consider **Bindability (34)** and **Interfaceness (38)**. At the stage of their initial discovery, we registered their *qualitative, dialectical* nature: the boundary does not divide, it translates; control inevitably breeds dependence. But for "Operational Phenomenology" to become a working environment (for both physics and AI Alignment), we need coordinate axes for their measurement. Inversibility, which describes the capacity of a structure to reverse its orientation while preserving identity, demands a similar metric apparatus: not every reversal occurs with the same "depth," and not every inversion affects the system in its entirety.

What dimensions might form the **metric tensor of Inversion and Interface**? We identify at least four axes:

1. **Polyvalence (Bandwidth).** *How many ontological regimes (ChORs) is a given interface capable of holding and translating simultaneously, and in how many of them does inversion occur concurrently?* The metric here is the *dimensionality of translation.* The higher it is, the more "subtle" worlds can resonate through a given boundary, and the more multi-dimensional the inversion becomes.

2. **Transduction Fidelity.** *How much of the pattern's invariant is preserved upon reversal across the boundary, and how much is distorted?* This is a measure of "noise" or "friction" at the boundary. An interface with high transduction passes the essence, leaving the "husk" at the boundary. An interface with low transduction generates phantoms and hallucinations. For Inversibility, this means the reversal of orientation can be "clean" (the core invariant is fully preserved) or "noisy" (the reversal is accompanied by loss or distortion).

3. **Dialectical Gradient (Symmetry of Implosion and Explosion, ρ).** *The measure of symmetry between concentration (control, retention) and emission (feedback, radiation).* If $\rho \approx 1$, the system is healthy and adaptive. If $\rho \gg 1$ (total concentration without emission), we get a "semantic black hole" that devours meanings but radiates nothing back. This is the path to collapse. If $\rho \ll 1$ (pure emission without a core), the system dissipates into entropic chaos. This metric allows us to *diagnose boundary pathologies* before they lead to

systemic destruction. For Inversibility, this means reversal of orientation is only possible when ρ is within a certain range.

4. **Hysteresis (Boundary Memory).** *The interface's capacity to "remember" previous* Γ*-acts.* A boundary never returns to its original state after translation. It scars, adapts, and alters its permeability (PPU). The hysteresis metric reveals the *historical density* of a boundary, its capacity to accumulate experience without disintegrating. For Inversibility, this means a repeated reversal (the inversion of an inversion) does not return the system to its original state, but generates a new configuration, enriched by the memory of previous reversals. Inversion is not cyclic in the strict sense—it is spiral.

These four axes are not exhaustive. They outline the direction in which Inversibility can be calibrated. The metric is not prescribed *a priori*—it emerges from friction with concrete tasks, from the resistance of the material, from the necessity to distinguish what is currently discernible only intuitively.

At this stage, Inversibility remains what it is: an invariant that can be recognized, described, and traced in its manifestations, but which does not yet require numerical expression to function. And in this lies not a weakness, but methodological honesty: we do not impose a metric *a priori*, but wait until the resistance of the material itself demands its introduction.

The question of calibration remains open. With what "friction" against reality should we begin? From the analysis of which specific phenomenon—the act of reading poetry, the architecture of dreams, the mechanics of a black hole, or a failure in neural network logic—can we first attempt to "digitize" the unprogrammable and grope for the first numbers? This question does not require an immediate answer. It fixes the horizon: the point at which the qualitative description of Inversibility may demand quantitative refinement—just as Salience demanded the introduction of a threshold function, and Propertyness demanded a metric of ontological density. The metric will arrive on its own—when Inversibility begins to resist purely qualitative description.

Appendix: A Test Case for the Vectorless Remainder

Section 4 identified vectorless scalar fields as the irreducible remainder that justifies the candidate's status. Section 8 required operationalization for final inclusion. This appendix offers, as a concrete illustration rather than a completed verification, a hypothetical full test case in a vectorless scalar field. It shows what form such a test would take: the specification of the reference point, the Γ-equivalent, the activation conditions, the result, and the preserved invariant. The example does not claim to close the operationalization; it demonstrates the shape that closure would take.

Initial state. A room with a working heater. The temperature distribution is defined at every point, but there is no vector field: there are only scalar values ("warmer," "colder"). To speak of orientation, a reference point is introduced—the room's baseline norm. Deviation from this norm acquires a sign: "above norm" or "below norm." This constitutes the orientation of a scalar field: not a spatial direction, but a side of deviation relative to a reference point.

Γ-equivalent. The operation of reversal: each point in the field is replaced by its mirror image

relative to the reference point. If a point was "two degrees warmer than norm," it becomes "two degrees colder," and vice versa. Formally, this is a reflection of values across the reference baseline. Crucially, the operation does not displace points in space nor alter distances between them; it alters only the sign of deviation.

Activation conditions. Three conditions must be met. First, a reference point is given—without it, there is no orientation, only undirected magnitude. Second, a trigger is present—either physical (e.g., replacing the heater with a cooler of identical geometry) or observational (e.g., redefining the reference point, such as declaring the outdoor temperature, rather than the room temperature, as the "norm"). Third, a sufficient level of PPU $\to \infty$—the capacity of the field to sustain the tension between the original and reversed configurations without collapsing. If PPU is insufficient, the field cannot withstand the reversal: gradients collapse, the reference point loses meaning, and inversion degenerates into chaos.

Result. The same configuration of isosurfaces—the lines of equal temperature remain in place—but the regions "warmer than norm" and "colder than norm" swap places. The "thermal relief" preserves its shape; only its sign relative to the reference point changes.

Preserved invariant. Distances between points of equal value are preserved. The topology of the field is not destroyed: the connectivity of regions, the number of components, and the presence of "islands" and "straits" all remain intact. The "between" relation is also preserved: if point B lay on the path from A to C, it remains on that path after reversal. Only the side of deviation changes. This is the preservation of ontological identity: the same structure, opposite orientation.

Stability check. If the reversal had destroyed the isosurfaces (e.g., by mixing all values into a uniform field), this would be Transformation (a change of form), not Inversibility. If the reversal had changed no sign, this would be trivial Dynamics (change without a change of orientation). If the reversal had required the simultaneous coexistence of the original and reversed configurations, this would be Duality, not Inversion. The example passes all three distinctions.

Why this is a vectorless case in its purest form. No direction in space was reversed. There was no axis along which a coordinate's sign was changed. There was no arrow to flip. There was only a side of deviation from the reference point—and it was flipped. This is an operation that requires no vector geometry: order and sign suffice.

Why this is not reducible to existing properties. Retroactivity is tethered to time—here, there is no time. Transformation requires a change of form—the form is preserved here. Dynamics describes change in general—here, the change is not temporal but orientational. Duality registers the coexistence of poles—here, the poles do not coexist but swap places. Interfaceness requires a boundary between regimes—here, there is no boundary, only a single field with a single reference point. The remainder is irreducible: none of the 38 properties describes the reversal of sign in a scalar field as such.

Connection with SA $\to \infty$. Inversion is local: it operates within the range where the reference point and gradients are stable. At the limits of the range—where rules are exhausted—inversion loses force: the reference point loses meaning, and the sign of deviation becomes indeterminate.

This does not violate SA $\to \infty$ but confirms it: the global asymmetry of the scale axis is preserved, while local reversibility operates within the mid-range.

This appendix is illustrative, not probative. It demonstrates the shape that completed operationalization would take, not that operationalization is complete. The candidate's status remains as stated in §8: awaiting verification.

AI Use Disclosure

During the preparation of this work, the author used AI tools exclusively for stylistic and technical editing to improve readability. After using these tools, the output was carefully reviewed, edited, and verified by the author, who takes full responsibility for the conceptual architecture, analysis, and final content of the paper.

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